

Key Classification of Music Using Chromagrams and CNNs

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Key Classification

The goal of this project was to build a model that could effectively identify the key of an input music clip



Music Theory - Minor Key

A natural minor scale (or Aeolian mode) is a diatonic scale that is built by starting on the sixth degree of its relative major scale.

What does this actually mean?



Why does the key matter?

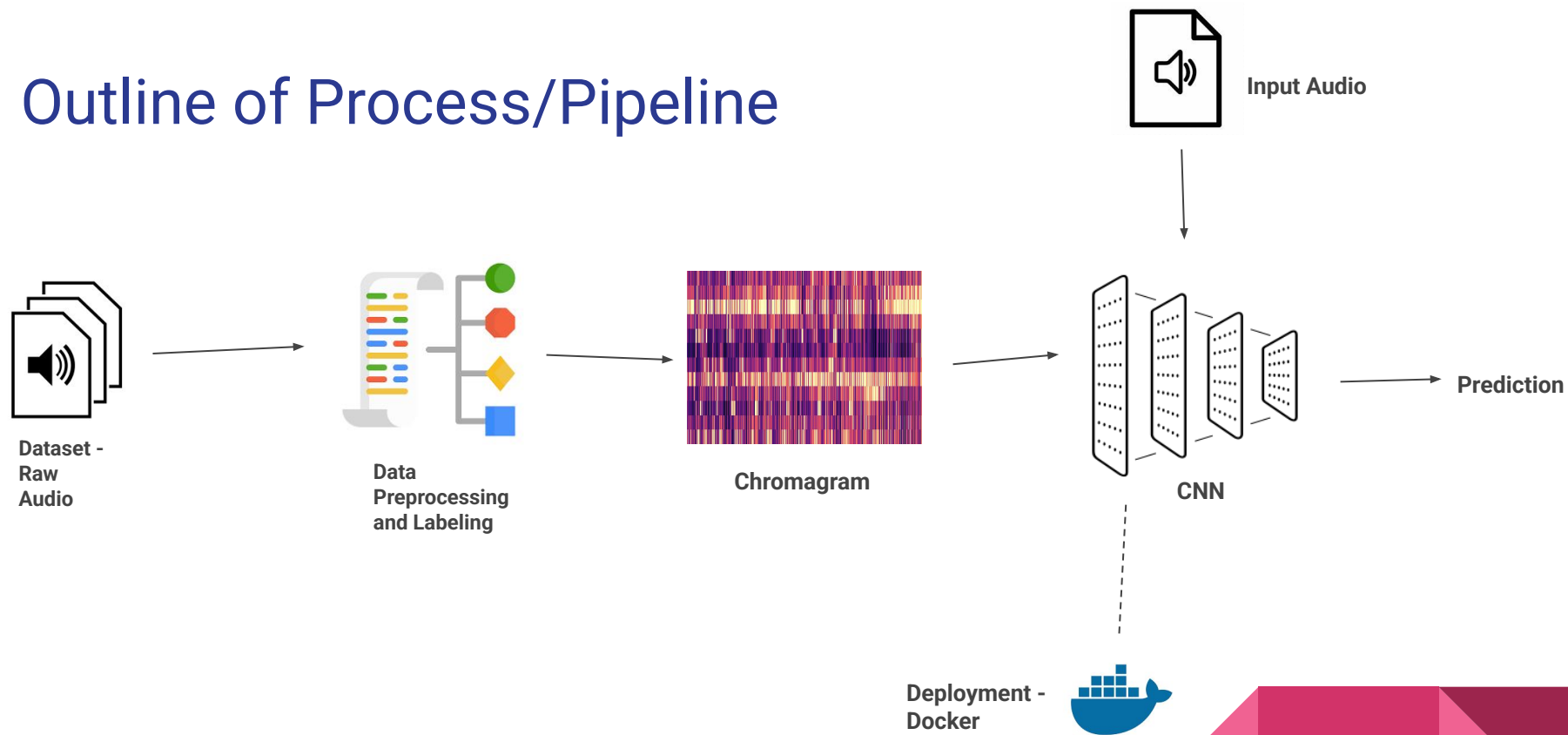
Sampling

Tuning Vocals

Creative Expansion



Outline of Process/Pipeline



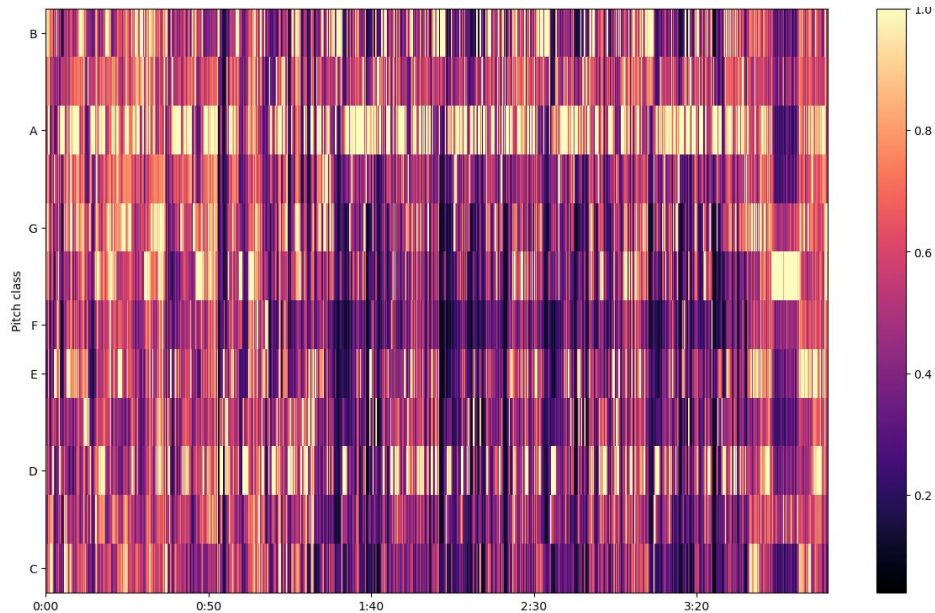
Spectrogram

A spectrogram is a visual representation of the spectrum of frequencies of a signal as it varies with time

What does this actually mean?



Chromagram



Dataset and Data Collection

Giantsteps Key Dataset

604 two minute long songs all 12 major and minor keys



Data Processing

Minor Key songs used for training

Labels 0 - 11 for each minor class

Librosa for Chromagrans

<https://blog.paperspace.com/audio-analysis-processing-maching-learning/>



Model and Training Model

Convolutional Neural Network

Model: "sequential"

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 222, 222, 32)	896
max_pooling2d (MaxPooling2D)	(None, 111, 111, 32)	0
conv2d_1 (Conv2D)	(None, 109, 109, 128)	36,992
max_pooling2d_1 (MaxPooling2D)	(None, 54, 54, 128)	0
conv2d_2 (Conv2D)	(None, 52, 52, 128)	147,584
max_pooling2d_2 (MaxPooling2D)	(None, 26, 26, 128)	0
conv2d_3 (Conv2D)	(None, 24, 24, 128)	147,584
max_pooling2d_3 (MaxPooling2D)	(None, 12, 12, 128)	0
flatten (Flatten)	(None, 18432)	0
dense (Dense)	(None, 12)	221,196

Total params: 554,252 (2.11 MB)

Trainable params: 554,252 (2.11 MB)

Non-trainable params: 0 (0.00 B)

```
!]: hist = model.fit(x_train_norm, y_train_encoded, validation_data=(x_test_norm, y_test_encoded), batch_size=10, epochs=10)
```

Deployment with Docker and FastAPI

Fast API Web Framework

Uvicorn

Docker



Demo



Challenges and Expansion

Data Collection

Music Theory - Circle of Fifths and Relative Major/Minor



